

Operations Research

Lecture 2: The Geometry and Algebra of Linear Programs

A textbook treatment

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October 1, 2026

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How to read this book

Each concept in this book is presented in the same rhythm.

- 1. Motivation and intuition.** Why do we care? What problem does the idea solve, and what picture should you hold in your head?
- 2. Formal statement.** The precise definition or theorem, set in a coloured box so it is easy to find when revising.
- 3. Proof.** Every claim is proved. Longer proofs are split into labelled steps, and the one idea that makes the proof work is isolated in a *Key insight* box.
- 4. Interpretation and discussion.** What the result really says, where it applies, and which misreadings to avoid.
- 5. Examples.** From “trivially small” to “genuinely useful.”
- 6. Techniques, classic problems and gotchas.** Computational shortcuts, proof patterns, exam favourites with one-line solutions, and the mistakes everyone makes once.

Margin notes carry the informal, “say it out loud” version of what is happening in the main text. Every major section closes with a summary box.

Prerequisites. Comfort with vectors, matrices, systems of linear equations, row reduction and the notions of rank and linear independence. Chapter 2 recalls exactly what is needed as it goes.

Chapter 2

The Geometry and Algebra of Linear Programs

2.1 From Scalars to Matrices

Motivation

Chapter 1 ended with a promise: linear programs can be solved exactly and efficiently. Delivering on that promise requires a change of viewpoint. Instead of a list of inequalities in $x_1, \dots, x_n$, we regard an LP as a statement about a matrix A , a right-hand side b and a cost vector c . Then we can ask linear-algebra questions (are these columns independent? is this system consistent? what is the rank?) whose answers turn out to locate the optimal solutions.

Matrix notation is not decoration: it lets us apply the whole toolbox of linear algebra (rank, independence, inverses) to a decision problem.

Recall from Chapter 1 that many OR problems have the shape

$$\max_{x \in X} f(x) \quad \text{subject to} \quad g_i(x) \leq 0 \quad (i = 1, \dots, m), \quad h_k(x) = 0 \quad (k = 1, \dots, p),$$

where $x = (x_1, \dots, x_n)$ collects the decisions and $X \subseteq \mathbb{R}^n$ specifies their domain. A linear program is the continuous case in which f , g_i and h_k are all linear.

Definition 2.1.1 (Canonical form, matrix notation). Let $A = (a_{ij}) \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$. The *canonical form* of an LP is

$$\max_{x \in \mathbb{R}^n} c^\top x \quad \text{subject to} \quad Ax \leq b, \quad x \geq 0,$$

where vector inequalities are understood *componentwise*: $Ax \leq b$ means $\sum_{j=1}^n a_{ij}x_j \leq b_i$ for every $i = 1, \dots, m$, and $x \geq 0$ means $x_j \geq 0$ for every j . Row i of A holds the coefficients of inequality i ; the numbers a_{ij}, b_i, c_j are data.

A dimension check keeps everyone honest: A is $m \times n$, x is $n \times 1$, so Ax is $m \times 1$, matching b . The objective $c^\top x$ is $(1 \times n)(n \times 1) = 1 \times 1$: a scalar, not a vector.

Proposition 2.1.2 (Componentwise inequalities behave well). For $u, v, w \in \mathbb{R}^m$ and $\lambda \geq 0$: if $u \leq v$ then $u + w \leq v + w$ and $\lambda u \leq \lambda v$; if $u \leq v$ and $y \geq 0$ then $y^\top u \leq y^\top v$.

Proof. Each statement holds coordinate by coordinate for real numbers; the last is $\sum_i y_i u_i \leq \sum_i y_i v_i$ because each term satisfies $y_i u_i \leq y_i v_i$ for $y_i \geq 0$. $\square$

Example 2.1.3 (Paint workshop, canonical form). A workshop makes paints A and B . Per litre, A needs 1 litre of base liquid and 2 g of pigment, earning 3 MAD; B needs 1 litre and 1 g, earning 2 MAD. Available: 4 litres of base, 6 g of pigment. Assumptions: fractional litres possible, constant recipes and profits, all production sold.

$$\max 3x_1 + 2x_2 \quad \text{s.t.} \quad x_1 + x_2 \leq 4, \quad 2x_1 + x_2 \leq 6, \quad x_1, x_2 \geq 0.$$

In matrix notation, with

$$G = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}, \quad h = \begin{bmatrix} 4 \\ 6 \end{bmatrix}, \quad p = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

the LP is $\max p^\top x$ s.t. $Gx \leq h, x \geq 0$: two decisions, two resource inequalities.

Summary: Matrix notation

- Canonical form: $\max c^\top x$ s.t. $Ax \leq b, x \geq 0$, inequalities componentwise.
- Check dimensions: $(m \times n)(n \times 1) = m \times 1$; $c^\top x$ is a scalar.
- Nonnegative multipliers preserve componentwise inequalities (Proposition 2.1.2); this is the algebraic heart of weak duality.

2.2 Equational Form

Motivation

Inequalities are awkward to manipulate algebraically: you cannot “solve” $x_1 + x_2 \leq 4$ for x_1 . Equations, by contrast, are the natural habitat of linear algebra. The equational form converts every inequality into an equation by giving the unused part of each resource its own name. The price is a few extra variables; the reward is that we can choose independent columns, solve a square system, and then simply *check* whether the result is nonnegative.

*Equalities let us solve;
nonnegativity lets us check.
The equational form
separates the two jobs.*

Definition 2.2.1 (Equational form). An LP is in *equational form* if it reads

$$\max_{z \in \mathbb{R}^n} c^\top z \quad \text{subject to} \quad Az = b, \quad z \geq 0,$$

with $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$: m equations and n variables, where n counts *all* variables including any that were added during the conversion. Its *feasible set* is $P = \{z \in \mathbb{R}^n : Az = b, z \geq 0\}$.

Slack: turn unused resources into variables

Example 2.2.2 (Slack in the paint workshop). For the plan $(x_1, x_2) = (1, 2)$, base liquid used is $x_1 + x_2 = 3$ of 4 litres: 1 litre is unused. Pigment used is $2x_1 + x_2 = 4$ of 6 g: 2 g unused. Call the unused amounts $s_1 = 1$ and $s_2 = 2$. In general

$$x_1 + x_2 + s_1 = 4, \quad s_1 \geq 0, \quad 2x_1 + x_2 + s_2 = 6, \quad s_2 \geq 0 :$$

used stock + unused stock = available stock. The equational form of Example 2.1.3 is

$$\begin{aligned} & \max \quad 3x_1 + 2x_2 + 0s_1 + 0s_2 \\ \text{s.t.} \quad & x_1 + x_2 + s_1 = 4, \quad 2x_1 + x_2 + s_2 = 6, \quad x_1, x_2, s_1, s_2 \geq 0, \end{aligned}$$

i.e. $\max c^\top z$ s.t. $Az = b, z \geq 0$ with

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix} \in \mathbb{R}^{2 \times 4}, \quad z = \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 6 \end{bmatrix}, \quad c = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 0 \end{bmatrix}.$$

The slack variables have objective coefficient 0 because profit depends only on production.

Definition 2.2.3 (Slack and surplus).

- *At most:* $a^\top x \leq b$ becomes $a^\top x + s = b, s \geq 0$. The *slack* $s = b - a^\top x$ is what is missing to reach capacity (a tank holding $q \leq 10$ litres: $s = 10 - q$).
- *At least:* $a^\top x \geq b$ becomes $a^\top x - s = b, s \geq 0$. The *surplus* $s = a^\top x - b$ is the amount beyond the minimum (an order of $q \geq 10$: $s = q - 10$).
- *Exactly:* $a^\top x = b$ stays as it is. No extra variable.

Gotcha

Never add a nonnegative variable to an *equality*. Requiring exactly $q = 10$ and writing $q + s = 10, s \geq 0$ would allow $q = 7, s = 3$; writing $q - s = 10$ would allow $q = 13$. Both change the requirement. Keep $a^\top x = b$ unchanged.

Theorem 2.2.4 (Canonical and equational forms are equivalent). *Let (C) be $\max\{c^\top x : Ax \leq b, x \geq 0\}$ and (E) be $\max\{c^\top x + 0^\top s : Ax + s = b, x \geq 0, s \geq 0\}$. The map $\phi(x) = (x, b - Ax)$ is a bijection from the feasible set of (C) onto the feasible set of (E) , with $c^\top x$ preserved. Consequently (C) and (E) have the same optimal value, x is optimal for (C) iff $\phi(x)$ is optimal for (E) , and (x, s) is optimal for (E) iff x is optimal for (C) .*

Proof. Step 1 (ϕ lands in the feasible set of (E)). If $Ax \leq b$ and $x \geq 0$, then $s := b - Ax \geq 0$ and $Ax + s = b$, so (x, s) is feasible for (E) .

Step 2 (injective). $\phi(x) = \phi(x')$ forces $x = x'$ (first component).

Step 3 (surjective). If (x, s) is feasible for (E) , then $s = b - Ax$ (from $Ax + s = b$) and $s \geq 0$ gives $Ax \leq b$; with $x \geq 0$, x is feasible for (C) and $\phi(x) = (x, s)$. So every feasible point of (E) has unique slacks determined by x , and dropping the slacks recovers a feasible x .

Step 4 (objective). $c^\top x + 0^\top s = c^\top x$. A bijection between feasible sets that preserves the objective maps optimal solutions to optimal solutions and preserves the optimal value. $\square$

Interpretation & discussion

Nothing about the decision has changed. Every feasible production plan has *unique* slacks, and the slacks carry useful information: a slack of 0 means the resource is fully used (the constraint is *tight* or *active*); a positive slack means spare capacity.

Make every variable nonnegative

Proposition 2.2.5 (Substitutions for bounded and free variables). *Each substitution below is a bijection (or, in the free case, a surjection) between the original domain of t and the new domain, so replacing t by the expression in u (and v) everywhere yields an LP with the same attainable objective values and the same optimal value.*

Original domain	Substitution	New domain
$t \geq L$	$t = L + u$	$u \geq 0$
$t \leq U$	$t = U - u$	$u \geq 0$
$L \leq t \leq U$	$t = L + u$	$0 \leq u \leq U - L$
$t \in \mathbb{R}$	$t = u - v$	$u, v \geq 0$

Proof. $u \mapsto L+u$ maps $[0, \infty)$ bijectively onto $[L, \infty)$ (inverse $u = t-L$); $u \mapsto U-u$ maps $[0, \infty)$ bijectively onto $(-\infty, U]$ (inverse $u = U-t$); $u \mapsto L+u$ maps $[0, U-L]$ bijectively onto $[L, U]$. The free case is Proposition 1.3.5. In each case the substitution is an identity in the constraints and objective, so feasible points correspond with equal objective values, and the sets of attainable values (hence the optimal values) coincide. $\square$

Example 2.2.6. For $-2 \leq t \leq 3$, use $t = u - 2$ with $0 \leq u \leq 5$. One shifted variable is enough; the upper bound $u \leq 5$ then becomes an ordinary $\leq$ constraint with its own slack. Use $u - v$ only for a genuinely free variable: it adds a variable and destroys uniqueness for nothing.

Technique: Converting to equational form

1. Make the objective a maximisation ($\min f \rightarrow \max -f$).
2. For every $\leq$ constraint add a slack $+s_i$; for every $\geq$ constraint subtract a surplus $-s_i$; leave equalities alone. Each new variable is ≥ 0 with objective coefficient 0.
3. For every variable not already ≥ 0 : shift ($t = L + u$), flip ($t = U - u$), or split ($t = u - v$) as in Proposition 2.2.5.
4. Recount: n is the total number of variables now; m is the number of equations. Write A with one column per variable and one row per equation, and c with a 0 for each added variable.
5. *Sanity checks:* slack columns of $\leq$ constraints form (part of) an identity matrix; every original feasible point must still be reachable; objective coefficients of slacks are 0.

Classic problems

- *Convert* $\min x_1 - x_2$ s.t. $x_1 + x_2 \geq 2$, $x_1 \leq 5$, $x_1 \geq 0$, $x_2 \in \mathbb{R}$. Set $x_2 = u - v$: $\max -x_1 + u - v$ s.t. $x_1 + u - v - s_1 = 2$, $x_1 + s_2 = 5$, $x_1, u, v, s_1, s_2 \geq 0$.
- *Read off slacks*. In the paint LP, at $(x_1, x_2) = (2, 2)$: $s_1 = 4 - 4 = 0$, $s_2 = 6 - 6 = 0$; both resources fully used.
- *How many variables?* A canonical LP with n variables and m constraints has $n + m$ variables in equational form.

Gotcha

- Adding a slack to an equality (see above).
- Forgetting to keep the bounds on a split variable: $-2 \leq t \leq 2$ with $t = u - v$ needs $u - v \leq 2$ and $-u + v \leq 2$ (each then gets its own slack).
- Giving the slack a nonzero objective coefficient. Unused stock earns nothing.
- Mixing up the sign: surplus is *subtracted*, slack is *added*.

Summary: Equational form

- $\max c^\top z$ s.t. $Az = b$, $z \geq 0$; feasible set $P = \{z : Az = b, z \geq 0\}$.
- Slack (+ s) for $\leq$, surplus ($-s$) for $\geq$, nothing for $=$; shift/flip/split variables to make them nonnegative.
- The conversion is an exact translation (Theorem 2.2.4, Proposition 2.2.5): same decisions, same optimal value.

2.3 A Linear Algebra Interlude: Solving $Az = b$

Motivation

Now that constraints are equations, the natural question is: what does the solution set of $Az = b$ look like, and how do we compute with it? Three facts are needed: row operations do not change the solution set; consistency can be read from the rank; and zero rows can be discarded. We prove them and then apply them to an LP with three variables and four constraints.

Definition 2.3.1 (Elementary row operations). On a matrix (typically the augmented matrix $[A \mid b]$): (i) swap two rows; (ii) multiply a row by a nonzero scalar; (iii) add a multiple of one row to another row. *Row reduction* (Gaussian elimination) applies these operations to reach a form in which each nonzero row starts with a *pivot* (its first nonzero entry) strictly to the right of the pivot of the row above, and all zero rows are at the bottom. The number of pivots is the *rank*.

Proposition 2.3.2 (Row operations preserve solution sets). *If $[A' \mid b']$ is obtained from $[A \mid b]$ by elementary row operations, then $\{z : Az = b\} = \{z : A'z = b'\}$.*

Proof. Each row of $[A | b]$ encodes one equation. Swapping equations, multiplying an equation by $\lambda \neq 0$, or adding a multiple of one equation to another produces equations that are consequences of the old ones, so every old solution satisfies the new system. Each operation is invertible by an operation of the same kind (swap back; multiply by $1/\lambda$; subtract the multiple), so every new solution also satisfies the old system. $\square$

Key insight

Invertibility. Each step can be undone, so nothing is lost or gained. The single most common error is to apply an operation to A but not to b ; the equation is then no longer the same equation.

Theorem 2.3.3 (Consistency). *The system $Az = b$ has a solution if and only if b lies in the column space $\text{col}(A) = \text{span}(a_1, \dots, a_n)$, if and only if $\text{rank}(A) = \text{rank}([A | b])$. Equivalently, after row reduction of $[A | b]$, there is no row of the form $[0 \cdots 0 | d]$ with $d \neq 0$.*

Proof. First equivalence. $Az = z_1 a_1 + \cdots + z_n a_n$ is a linear combination of the columns, so a solution exists iff b is such a combination, i.e. $b \in \text{col}(A)$.

Second equivalence. We have $\text{rank}([A | b]) = \dim \text{span}(a_1, \dots, a_n, b)$. If $b \in \text{span}(a_1, \dots, a_n)$, adding b does not enlarge the span, so the ranks are equal. If $b \notin \text{span}(a_1, \dots, a_n)$, adding b increases the dimension by one, so $\text{rank}([A | b]) = \text{rank}(A) + 1$.

Row form. Row reduction preserves the rank of A and of $[A | b]$ (row operations do not change row spaces). A reduced row $[0 \cdots 0 | d]$ with $d \neq 0$ is a pivot in the last column, which happens exactly when $\text{rank}([A | b]) > \text{rank}(A)$; and such a row reads $0 = d$, which no z satisfies. $\square$

Proposition 2.3.4 (Zero rows may be deleted). *If the reduced augmented matrix contains a row $[0 \cdots 0 | 0]$, deleting it does not change the solution set. After deleting all such rows, the remaining matrix $\tilde{A} \in \mathbb{R}^{r \times n}$ has $r = \text{rank}(A)$ rows, which are linearly independent, and $\{z : \tilde{A}z = \tilde{b}\} = \{z : Az = b\}$.*

Proof. The row $0 = 0$ holds for every z , so removing it changes nothing. The remaining rows are the nonzero rows of a row-reduced matrix; each has a pivot in a distinct column, so they are independent, and their number equals the number of pivots, i.e. the rank. The equality of solution sets is Proposition 2.3.2 plus the deletion just justified. $\square$

Redundant inequalities

Before adding slacks, it is wise to discard inequalities that are implied by others: they only add variables and equations without changing the feasible set.

Proposition 2.3.5 (Nonnegative combinations are redundant). *Suppose inequality k satisfies $a_k = \sum_{i \neq k} \lambda_i a_i$ and $b_k \geq \sum_{i \neq k} \lambda_i b_i$ for some $\lambda_i \geq 0$. Then every x satisfying the other inequalities $a_i^\top x \leq b_i$ ($i \neq k$) also satisfies $a_k^\top x \leq b_k$; the feasible set is unchanged when inequality k is deleted.*

Proof. $a_k^\top x = \sum_{i \neq k} \lambda_i a_i^\top x \leq \sum_{i \neq k} \lambda_i b_i \leq b_k$, using $\lambda_i \geq 0$ for the first inequality. $\square$

Worked example: build a basic feasible solution by hand**Example 2.3.6** (A three-variable LP).

$$\begin{aligned} & \max \quad 3x_1 + 2x_2 + x_3 \quad \text{s.t.} \quad \begin{aligned} & x_1 + x_2 + x_3 \leq 4, \\ & 2x_1 + 3x_2 + x_3 \leq 10, \\ & 3x_1 + 4x_2 + 2x_3 \leq 14, \\ & 3x_1 + 5x_2 + x_3 \leq 18, \end{aligned} \quad x_1, x_2, x_3 \geq 0. \end{aligned}$$

Our goal: construct a basic feasible solution. We simplify, add slacks, eliminate, select variables, and check.

Step 1: remove a redundant inequality. Adding the first two inequalities gives $3x_1 + 4x_2 + 2x_3 \leq 14$, which is exactly the third. By Proposition 2.3.5 (with $\lambda_1 = \lambda_2 = 1$), the third inequality is implied and we remove it before adding slacks. (The fourth is also redundant: $\frac{1}{2}(\text{first}) + \frac{3}{2}(\text{second})$ gives $3.5x_1 + 5x_2 + 2x_3 \leq 17$, which implies the fourth for $x \geq 0$; we keep it anyway, to practise with three equations.)

Step 2: add three slacks.

$$x_1 + x_2 + x_3 + s_1 = 4, \quad 2x_1 + 3x_2 + x_3 + s_2 = 10, \quad 3x_1 + 5x_2 + x_3 + s_3 = 18,$$

with $z = (x_1, x_2, x_3, s_1, s_2, s_3)^\top \geq 0$ and $c = (3, 2, 1, 0, 0, 0)^\top$; $A \in \mathbb{R}^{3 \times 6}$.

Step 3: row-reduce $[A \mid b]$, columns in the order $x_1, x_2, x_3, s_1, s_2, s_3 \mid b$:

$$\begin{aligned} [A \mid b] &= \left[\begin{array}{cccccc|c} 1 & 1 & 1 & 1 & 0 & 0 & 4 \\ 2 & 3 & 1 & 0 & 1 & 0 & 10 \\ 3 & 5 & 1 & 0 & 0 & 1 & 18 \end{array} \right] \xrightarrow{\substack{R_2 \leftarrow R_2 - 2R_1 \\ R_3 \leftarrow R_3 - 3R_1}} \left[\begin{array}{cccccc|c} 1 & 1 & 1 & 1 & 0 & 0 & 4 \\ 0 & 1 & -1 & -2 & 1 & 0 & 2 \\ 0 & 2 & -2 & -3 & 0 & 1 & 6 \end{array} \right] \\ &\xrightarrow{R_3 \leftarrow R_3 - 2R_2} \left[\begin{array}{cccccc|c} 1 & 1 & 1 & 1 & 0 & 0 & 4 \\ 0 & 1 & -1 & -2 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -2 & 1 & 2 \end{array} \right]. \end{aligned}$$

Three pivots (columns 1, 2, 4): $\text{rank}(A) = \text{rank}([A \mid b]) = 3$. The system is consistent with three independent rows, and there is no zero row to delete: each retained inequality has its own slack, and the three slack columns form an identity matrix in the original A , which already guarantees full row rank. The equivalent LP is

$$\begin{aligned} & \max \quad 3x_1 + 2x_2 + x_3 \quad \text{s.t.} \quad \begin{aligned} & x_1 + x_2 + x_3 + s_1 = 4, \\ & x_2 - x_3 - 2s_1 + s_2 = 2, \quad z \geq 0. \\ & s_1 - 2s_2 + s_3 = 2, \end{aligned} \end{aligned}$$

Step 4: choose three of the six variables, whose columns must be independent; set the other three to zero; solve; check signs.

First choice: x_1, x_2, x_3 , with $s_1 = s_2 = s_3 = 0$. The reduced system restricted to these columns is

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 2 \end{array} \right].$$

Only two pivots: the three selected columns are dependent, and the last row reads $0 = 2$. This choice gives no solution at all. Reject it; the LP itself may still be feasible.

Second choice: x_1, x_2, s_3 , with $x_3 = s_1 = s_2 = 0$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{array} \right] \implies s_3 = 2, x_2 = 2, x_1 = 4 - x_2 = 2.$$

Three pivots: independent columns. The solution $z = (2, 2, 0, 0, 0, 2)^\top \geq 0$ is a *basic feasible solution*.

Check in the canonical LP. For $x = (2, 2, 0)$ all four original inequalities hold: $4 \leq 4$, $10 \leq 10$, $14 \leq 14$, $16 \leq 18$. The slacks of the three retained constraints are $s_1 = 0$, $s_2 = 0$, $s_3 = 2$, matching the computation; the removed inequality is automatically satisfied (with equality, as it must be, being the sum of two tight ones). Objective: $3(2) + 2(2) + 0 = 10$.

Third choice: x_1, x_3, s_3 , with $x_2 = s_1 = s_2 = 0$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 0 & -1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{array} \right] \implies s_3 = 2, x_3 = -2, x_1 = 6.$$

Three pivots again: independent columns, and the system has a unique solution. But $x_3 = -2 < 0$: a basic solution that is *not feasible*. Nonnegativity is a separate check *after* solving.

Interpretation & discussion

Three lessons from one example. (1) Independence of the chosen columns is necessary to get a solution at all. (2) Independence is not sufficient for feasibility: the sign check is separate. (3) A basic feasible solution is one *candidate*, not the optimum. Indeed $x = (4, 0, 0)$ is feasible ($4 \leq 4$, $8 \leq 10$, $12 \leq 18$) with objective $12 > 10$; it corresponds to the choice x_1, s_2, s_3 giving $z = (4, 0, 0, 0, 2, 6)$. And 12 is in fact optimal: by Lemma 1.3.10 with $y = (3, 0, 0)$ on the three retained constraints, $A^\top y = (3, 3, 3) \geq (3, 2, 1) = c$ and $b^\top y = 12$, so no feasible point exceeds 12.

Technique: The general recipe: reduce, then build a candidate

Given $\max c^\top z$ s.t. $Az = b$, $z \geq 0$ with $A \in \mathbb{R}^{m \times n}$:

1. **Row-reduce** $[A | b]$. Apply every operation to the right-hand side too.
2. **Check consistency.** A row $[0 \cdots 0 | d]$ with $d \neq 0$ means no solution: the LP is infeasible (Theorem 2.3.3).
3. **Delete zero rows.** With $r = \text{rank}(A)$, the equivalent system $\tilde{A}z = \tilde{b}$ has $\tilde{A} \in \mathbb{R}^{r \times n}$ of rank r (Proposition 2.3.4). Feasible set and objective are unchanged.
4. **Choose r independent columns.** $B \subseteq \{1, \dots, n\}$, $|B| = r$, with $\tilde{A}_B$ invertible; $N = \{1, \dots, n\} \setminus B$.
5. **Set nonbasic variables to zero; solve for the rest.** $z_N = 0$, $\tilde{A}_B z_B = \tilde{b}$.
6. **Check nonnegativity.** If $z_B \geq 0$ the result is a BFS; otherwise this choice gives an infeasible basic solution. One choice gives one candidate,

not necessarily an optimum.

Gotcha

- Forgetting to transform b along with A .
- Confusing “this choice of columns has no solution” with “the LP is infeasible.” Only a row $[0 \cdots 0 \mid d \neq 0]$ in the reduction of the *full* $[A \mid b]$ proves infeasibility.
- Skipping the sign check. Solving the square system is half the job.
- Removing an inequality that is *not* implied (e.g. one obtained by subtracting others). Only nonnegative combinations are safe.

Summary: Solving $Az = b$

- Row operations are invertible and preserve the solution set; do them on $[A \mid b]$.
- Consistent iff $b \in \text{col}(A)$ iff $\text{rank } A = \text{rank}[A \mid b]$ iff no row $[0 \cdots 0 \mid d \neq 0]$.
- Delete zero rows to reach full row rank r ; then choose r independent columns, set the others to zero, solve, check signs.
- An inequality that is a nonnegative combination of others is redundant; drop it before adding slacks.

2.4 Bases and Basic Feasible Solutions

Motivation

The feasible set of an LP usually contains infinitely many points, so “compare them all” is not an algorithm. The worked example suggested a way out: certain feasible points are obtained by fixing all but a few variables to zero and solving a square system. There are only finitely many ways to choose the square system. If these special points suffice to find the optimum, we have a finite algorithm. This section defines them algebraically, connects them to the geometric idea of a corner, and works through every basis of a small example.

Infinitely many feasible points, but only finitely many corners. If we can prove the optimum sits at a corner, the search becomes finite.

Example 2.4.1 (Sharing network resources). A server handles traffic for two services with throughputs $x_1, x_2 \geq 0$ (Gbit/s). Per Gbit/s, service A uses 1 CPU unit and 2 bandwidth units; service B uses 1 and 1. Capacities: 6 CPU units, 8 bandwidth units:

$$x_1 + x_2 \leq 6, \quad 2x_1 + x_2 \leq 8.$$

With slacks $s_1 = 6 - x_1 - x_2$ (unused CPU) and $s_2 = 8 - 2x_1 - x_2$ (unused bandwidth), the equational feasible set is

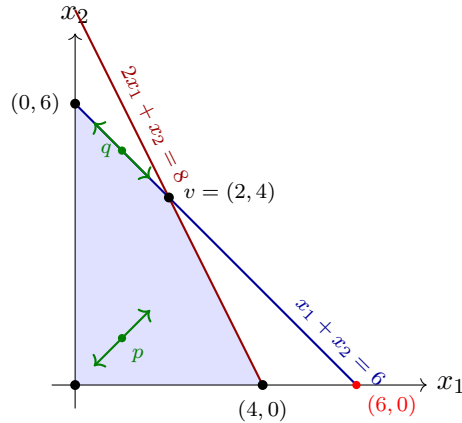
$$P = \{z = (x_1, x_2, s_1, s_2) \geq 0 : x_1 + x_2 + s_1 = 6, 2x_1 + x_2 + s_2 = 8\},$$

that is, $Az = b, z \geq 0$ with

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 6 \\ 8 \end{bmatrix}.$$

A feasible solution is any point of P ; it need not be optimal. For instance $z = (1, 1, 4, 5)$ satisfies both equations and is nonnegative. We could maximise total throughput $x_1 + x_2$; for now we study feasibility and bases.

Geometric intuition



At p (interior) or q (on an edge) we can move a little in *both* directions along some straight line and remain feasible. At a vertex such as $v = (2, 4)$, no such two-sided movement is possible: every line through v leaves the region on at least one side. The picture uses (x_1, x_2) ; the complete vector also contains the slacks.

Definition 2.4.2 (Extreme point). A point z of a convex set P is an *extreme point* (or *vertex*) of P if it is not the midpoint of two distinct points of P : there do not exist $u \neq w$ in P with $z = \frac{1}{2}(u + w)$. Equivalently, z cannot be written as $\lambda u + (1 - \lambda)w$ with $u \neq w$ in P and $0 < \lambda < 1$.

(The two formulations agree: if $z = \lambda u + (1 - \lambda)w$ with $0 < \lambda < 1$, $u \neq w$, then replacing u, w by suitable points on the segment $[u, w]$ symmetric about z writes z as a midpoint; the converse is the case $\lambda = \frac{1}{2}$.)

The algebraic definition

Definition 2.4.3 (Basis, basic solution, basic feasible solution). Let $A \in \mathbb{R}^{m \times n}$ have $\text{rank}(A) = m$ (full row rank), and write $A = (a_1 \cdots a_n)$ by columns. A *basis* is an index set $B \subseteq \{1, \dots, n\}$ with $|B| = m$ such that the square matrix $A_B = (a_j)_{j \in B}$ is invertible. Let $N = \{1, \dots, n\} \setminus B$ (the *nonbasic* indices) and write $z_B = (z_j)_{j \in B}$, $z_N = (z_j)_{j \in N}$. The *basic solution* associated with B is the unique vector z with

$$z_N = 0, \quad z_B = A_B^{-1}b.$$

It is a *basic feasible solution* (BFS) if in addition $z_B \geq 0$. A feasible vector $z \in P$ is called *basic* if it is the basic solution of at least one basis. The variables z_j , $j \in B$, are the *basic variables*.

Proposition 2.4.4 (Basic solutions are well defined and satisfy $Az = b$). For a basis B , the basic solution exists, is unique, and satisfies $Az = b$. It has at most m nonzero entries.

Proof. The equations split as $Az = A_B z_B + A_N z_N = b$. Imposing $z_N = 0$ leaves $A_B z_B = b$, a square system with invertible matrix, so $z_B = A_B^{-1}b$ is its unique solution. Substituting back, $Az = A_B A_B^{-1}b + A_N 0 = b$. Only the m entries in z_B can be nonzero. $\square$

Remark 2.4.5 (Degeneracy). Basic variables may themselves equal zero: a BFS has *at most* m positive entries, not necessarily exactly m . Such a BFS is called *degenerate*, and it typically arises from several different bases. For example, with constraints $x_1 \leq 1$, $x_1 + x_2 \leq 1$ (so $A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$, $b = (1, 1)^\top$), the bases $\{x_1, x_2\}$, $\{x_1, s_1\}$ and $\{x_1, s_2\}$ all produce the same point $z = (1, 0, 0, 0)$.

Every basis of the network example

Example 2.4.6 (Basis $\{1, 2\}$: both throughputs). Choose $B = \{1, 2\}$, $N = \{3, 4\}$. $A_B = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$ with $\det A_B = -1 \neq 0$: a basis. Set $s_1 = s_2 = 0$ and solve $x_1 + x_2 = 6$, $2x_1 + x_2 = 8$: $x_1 = 2$, $x_2 = 4$. So $z = (2, 4, 0, 0)^\top \geq 0$ is a BFS. It is the corner where both resources are fully used.

Example 2.4.7 (Basis $\{1, 3\}$: a different corner). $B = \{1, 3\}$ (solve for x_1 and s_1), $A_B = \begin{bmatrix} 1 & 1 \\ 2 & 0 \end{bmatrix}$, $\det A_B = -2 \neq 0$. Set $x_2 = s_2 = 0$: $2x_1 = 8$ gives $x_1 = 4$, then $s_1 = 6 - 4 = 2$. So $z = (4, 0, 2, 0)^\top \geq 0$: another BFS. No service- B traffic; bandwidth fully used; two CPU units remain. This is the corner $(4, 0)$.

Example 2.4.8 (Basis $\{1, 4\}$: a basis that produces an infeasible point). $B = \{1, 4\}$ selects x_1, s_2 ; set $x_2 = s_1 = 0$. Then $x_1 = 6$ and $2x_1 + s_2 = 8$ gives $s_2 = -4$. $A_B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ is nonsingular, so B is a basis, but $s_2 = -4 < 0$: not a feasible basis. Geometrically, $(6, 0)$ is the intersection of $x_2 = 0$ and $x_1 + x_2 = 6$ but violates $2x_1 + x_2 \leq 8$ (the red point in the figure).

Example 2.4.9 (All bases: finite candidates). There are $\binom{4}{2} = 6$ two-column sets.

Basis B	Basic solution (x_1, x_2, s_1, s_2)	Status
$\{3, 4\}$	$(0, 0, 6, 8)$	feasible
$\{1, 3\}$	$(4, 0, 2, 0)$	feasible
$\{1, 2\}$	$(2, 4, 0, 0)$	feasible
$\{2, 4\}$	$(0, 6, 0, 2)$	feasible
$\{1, 4\}$	$(6, 0, 0, -4)$	infeasible
$\{2, 3\}$	$(0, 8, -2, 0)$	infeasible

(Every pair of columns here is independent, so all six are bases.) Four feasible bases, four vertices of the original polygon: $(0, 0)$, $(4, 0)$, $(2, 4)$, $(0, 6)$. The two infeasible basic solutions are intersections of two constraint lines that fall outside the polygon.

Vertices are exactly the basic feasible solutions

Theorem 2.4.10 (BFS = extreme point). *Let $A \in \mathbb{R}^{m \times n}$ have rank m and $P = \{z : Az = b, z \geq 0\}$. A point $z \in P$ is a basic feasible solution if and only if it is an extreme point of P .*

Proof. ($\Rightarrow$) **A BFS is extreme.** Let z be the BFS of a basis B , and suppose $z = \lambda u + (1 - \lambda)w$ with $u, w \in P$ and $0 < \lambda < 1$. For $j \in N$: $0 = z_j = \lambda u_j + (1 - \lambda)w_j$ with $u_j, w_j \geq 0$ and both coefficients positive, so $u_j = w_j = 0$. Thus $u_N = w_N = 0$, and $Au = b$ reduces to $A_B u_B = b$; likewise $A_B w_B = b$. Since A_B is invertible, $u_B = w_B = A_B^{-1}b = z_B$. Hence $u = w = z$, so z is not a proper combination of two distinct points: it is extreme.

($\Leftarrow$) **An extreme point is a BFS.** Let $z \in P$ be extreme and let $S = \{j : z_j > 0\}$ be its support.

Step 1: the columns $\{a_j : j \in S\}$ are linearly independent. Suppose not: there is $d \neq 0$ with $d_j = 0$ for $j \notin S$ and $Ad = \sum_{j \in S} d_j a_j = 0$. For $\varepsilon > 0$, the points $z \pm \varepsilon d$ satisfy $A(z \pm \varepsilon d) = b$. Their entries outside S are 0, and inside S they are $z_j \pm \varepsilon d_j$, which are positive for ε small because $z_j > 0$. So $z + \varepsilon d$ and $z - \varepsilon d$ are two distinct points of P with midpoint z , contradicting extremality.

Step 2: extend to a basis. The independent set $\{a_j : j \in S\}$ has at most m elements. Since $\text{rank } A = m$, the n columns of A span $\mathbb{R}^m$, so any independent subset of columns can be extended, using further columns of A , to a set of m independent columns, i.e. to a basis $B \supseteq S$. (Repeatedly add a column not in the span of the current set; this is possible until the span is all of $\mathbb{R}^m$, which happens exactly when m independent columns have been collected.)

Step 3: z is the basic solution of B . For $j \notin B$ we have $j \notin S$, so $z_j = 0$: $z_N = 0$. Then $Az = b$ reads $A_B z_B = b$, so $z_B = A_B^{-1}b$. Thus z is the basic solution of B , and $z \geq 0$ makes it a BFS. $\square$

Key insight

The direction d in Step 1 is a *feasible direction of two-sided movement*: because $Ad = 0$, moving along $\pm d$ keeps the equations, and because d lives only on coordinates where z is strictly positive, small moves keep nonnegativity. A point admits such a d exactly when the columns under its support are dependent. So “no two-sided wiggle room” (geometry) is the same as “independent support columns” (algebra).

Interpretation & discussion

The theorem is why we may say “vertex” and “basic feasible solution” interchangeably. It also explains the table above: the four feasible bases produced exactly the four corners of the polygon, and no corner was missed. Note the role of the hypothesis $\text{rank } A = m$: without it, an independent support might not extend to m columns; this is why Section 2.3 insisted on deleting redundant equations first.

Corollary 2.4.11 (Finitely many vertices). *P has at most $\binom{n}{m}$ extreme points.*

Proof. Each extreme point is the basic solution of some basis (Theorem 2.4.10), each basis is a subset of size m of $\{1, \dots, n\}$, and each basis determines exactly one basic solution (Proposition 2.4.4). There are $\binom{n}{m}$ such subsets. $\square$

Technique: Working with bases by hand

- *Two-by-two invertibility*: $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc \neq 0$. Larger: row-reduce A_B and count pivots (m pivots means invertible).
- *Identity columns are free*: if the slack columns form an identity matrix, then $B = \{\text{all slacks}\}$ is always a basis and its basic solution is $x = 0$, $s = b$; it is feasible iff $b \geq 0$ (the origin of the polygon).
- *Read the corner*: a basis selects which constraints are *tight*. Nonbasic slack $s_i = 0$ means constraint i is tight; nonbasic $x_j = 0$ means the axis $x_j = 0$ is tight. Two tight lines in the plane meet at the corner.
- *Sanity check a BFS*: plug x back into the *original* inequalities. If any fails, the corresponding slack is negative and you mis-solved.
- *Count*: at most $\binom{n}{m}$ bases; in a 2-constraint, 4-variable problem that is 6, and you can list them all.

Classic problems

- Which point does basis $\{2, 4\}$ of Example 2.4.1 represent? $x_1 = s_1 = 0$, $x_2 = 6$, $s_2 = 8 - 6 = 2$: the corner $(0, 6)$; CPU fully used, 2 bandwidth units spare.
- Is $(1, 1, 4, 5)$ a BFS of Example 2.4.1? No: it has four positive entries but $m = 2$.
- Is $(3, 3, 0, -1)$ basic? It is the basic solution of $B = \{1, 2\}$ only if it solves $A_B z_B = b$; but $B = \{1, 2\}$ gives $(2, 4)$. In fact $(3, 3, 0, -1)$ is not even feasible ($s_2 < 0$), so it is not a BFS.
- A point with m positive entries whose columns are independent is a BFS. Yes: its support is a basis and it solves $A_B z_B = b$ with $z_B > 0$.

Gotcha

- *Basis $\neq$ feasible basis*. Invertibility of A_B says nothing about signs.
- *Not every m -subset is a basis*. Columns can be dependent (x_1, x_2, x_3 in Example 2.3.6).
- *Degenerate BFS*. One point, several bases. Do not conclude “different basis, different corner.”
- *Full row rank is required*. If $\text{rank } A < m$, no $m \times m$ submatrix is invertible and Definition 2.4.3 is vacuous; reduce first.

Summary: Bases and BFSs

- For A of full row rank m : a basis is m independent columns; set the other variables to 0 and solve $A_B z_B = b$; a BFS additionally has $z_B \geq 0$.
- BFSs are exactly the extreme points (vertices) of P (Theorem 2.4.10); there are at most $\binom{n}{m}$ of them.

- A BFS has at most m positive entries; fewer means degenerate.

2.5 The Fundamental Theorem of Linear Programming

Motivation

We now have a finite list of candidates: the basic feasible solutions. The question is whether the optimum is guaranteed to be among them. The answer is yes, under the two hypotheses one would expect (the problem is feasible, and the objective cannot grow without bound), and the proof is constructive: it shows how to walk from any feasible point to a vertex that is at least as good.

If a feasible point is not a vertex, slide along a feasible direction without losing objective value until a coordinate hits zero. Repeat. You arrive at a vertex at least as good.

Lemma 2.5.1 (Purification). *Let $A \in \mathbb{R}^{m \times n}$ have rank m , let $P = \{z : Az = b, z \geq 0\}$, and suppose $c^\top z$ is bounded above on P . Then for every $z \in P$ there is a basic feasible solution z' with $c^\top z' \geq c^\top z$.*

Proof. We argue by induction on the number $k = |\text{supp}(z)|$ of positive entries of z , where $\text{supp}(z) = \{j : z_j > 0\}$.

Base and “independent” case. If the columns $\{a_j : j \in \text{supp}(z)\}$ are linearly independent (this includes $k = 0$, the empty set being independent), then exactly as in Steps 2–3 of the proof of Theorem 2.4.10, $\text{supp}(z)$ extends to a basis B and z is the BFS of B . Take $z' = z$.

Dependent case. Otherwise there is $d \neq 0$ with $\text{supp}(d) \subseteq \text{supp}(z)$ and $Ad = 0$. Replacing d by $-d$ if necessary, we may assume $c^\top d \geq 0$. We claim we may also assume d has at least one negative entry:

- if $c^\top d > 0$ and $d \geq 0$, then $z + \theta d \in P$ for all $\theta \geq 0$ (equations hold since $Ad = 0$; nonnegativity since $d \geq 0$) and $c^\top(z + \theta d) = c^\top z + \theta c^\top d \rightarrow \infty$, contradicting boundedness; so some $d_j < 0$;
- if $c^\top d = 0$ and $d \geq 0$, replace d by $-d$: still $Ad = 0$, $c^\top d = 0$, and now some entry is negative since $d \neq 0$.

Now define

$$\theta^* = \min \left\{ \frac{z_j}{-d_j} : d_j < 0 \right\} > 0,$$

positive because every j with $d_j < 0$ lies in $\text{supp}(z)$, where $z_j > 0$. Set $z' = z + \theta^* d$. Then:

- $Az' = b$, since $Ad = 0$;
- $z' \geq 0$: for $d_j \geq 0$, $z'_j \geq z_j \geq 0$; for $d_j < 0$, $z'_j = z_j + \theta^* d_j \geq z_j - \frac{z_j}{-d_j}(-d_j) = 0$ by the choice of θ^* ;
- $c^\top z' = c^\top z + \theta^* c^\top d \geq c^\top z$;
- $|\text{supp}(z')| < |\text{supp}(z)|$: no new positive entries appear (outside $\text{supp}(z)$ both z and d vanish), and for the index j attaining the minimum, $z'_j = 0$.

By the induction hypothesis applied to z' there is a BFS z'' with $c^\top z'' \geq c^\top z' \geq c^\top z$. $\square$

Key insight

Two ingredients. *Dependence* gives a direction d with $Ad = 0$ supported where z is positive, so moving along it keeps the equations and, for a while, the signs. *Boundedness forbids* an improving direction that never hits a bound; therefore some coordinate decreases, and the *ratio test* $\theta^* = \min z_j/(-d_j)$ tells us exactly how far to go before the first coordinate reaches zero. The ratio test will reappear as the heart of the simplex method.

Theorem 2.5.2 (Existence of an optimal BFS). *Let $A \in \mathbb{R}^{m \times n}$ have full row rank (after removing redundant equations) and consider*

$$\max c^\top z \quad \text{subject to} \quad Az = b, \quad z \geq 0.$$

If the feasible set P is nonempty and the objective is bounded above on P , then the problem has an optimal solution, and at least one optimal solution is a basic feasible solution.

Proof. **Step 1 (there is at least one BFS).** $P \neq \emptyset$; pick any $z \in P$ and apply Lemma 2.5.1 to obtain a BFS. So the set $\mathcal{B}$ of basic feasible solutions is nonempty, and it is finite by Corollary 2.4.11.

Step 2 (the best BFS beats every feasible point). Let $z^* \in \mathcal{B}$ maximise $c^\top z$ over the finite set $\mathcal{B}$ (a maximum over a nonempty finite set exists). For any $z \in P$, Lemma 2.5.1 gives a BFS z' with $c^\top z \leq c^\top z' \leq c^\top z^*$.

Step 3 (conclusion). $z^* \in P$ and $c^\top z^* \geq c^\top z$ for all $z \in P$: z^* is an optimal solution, and it is a BFS. $\square$

Interpretation & discussion

- *Finding the best BFS gives the optimal value.* The infinite search over P collapses to a finite comparison. This is the licence for every vertex-based algorithm, the simplex method first among them.
- *Other optimal solutions may exist that are not basic.* If two vertices are optimal, the whole edge between them is optimal (Proposition 1.7.9), and its interior points are not BFSs. The theorem promises *at least one* optimal vertex, not that all optima are vertices.
- *The hypotheses are exactly the two bad outcomes.* Infeasible: no BFS exists at all. Unbounded: the purification argument breaks precisely because an improving ray $z + \theta d$ exists. So for LPs, “feasible and bounded” already implies “solvable,” closing the gap noted after Definition 1.2.4.

Example 2.5.3 (The network example, optimised). Maximise total throughput $x_1 + x_2$ over Example 2.4.1. The four BFSs have values 0 at $(0,0)$, 4 at $(4,0)$, 6 at $(2,4)$ and 6 at $(0,6)$. By Theorem 2.5.2 the optimal value is 6. Both $(2,4)$ and $(0,6)$ are optimal, and so is every point of the segment between them, e.g. $(1,5)$ with slacks $(0,2)$: an optimal solution with three positive entries, hence not basic. Certificate: $x_1 + x_2 \leq 6$ is itself a constraint, so $y = (1,0)$ proves the bound.

A first algorithm: compare the bases

Technique: Complete enumeration of bases

Assume A has m independent rows and the LP is feasible with bounded objective.

1. For each set B of m column indices:
2. if A_B is invertible, solve $A_B z_B = b$ and set $z_N = 0$;
3. keep the solutions with $z_B \geq 0$;
4. compare $c^\top z$ over the kept solutions and return the best.

Theorem 2.5.2 guarantees the answer is optimal. The method is practical for small examples: $\binom{4}{2} = 6$ systems above. It is hopeless at scale: an LP with $m = 50$ equations and $n = 100$ variables has $\binom{100}{50} \approx 10^{29}$ candidate bases. The simplex method, next in the course, walks from BFS to neighbouring BFS improving the objective at each step, visiting only a tiny fraction of them.

Classic problems

- “An LP is feasible and bounded. Must it have an optimal BFS?” Yes, after reducing to full row rank (Theorem 2.5.2).
- “An LP has an optimal solution that is not a vertex. Contradiction?” No: it then has infinitely many optimal solutions, at least one of which is a vertex.
- “The feasible set is $\{x \geq 0 : x_1 - x_2 = 0\}$ and we maximise x_1 .” Unbounded: $d = (1, 1)$ has $Ad = 0$, $d \geq 0$, $c^\top d = 1 > 0$. This is exactly the ray the purification lemma detects.
- Purify $(1, 1, 4, 5)$ in Example 2.4.1 for $c = (1, 1, 0, 0)$. Support is everything; four columns in $\mathbb{R}^2$ are dependent. Take $d = (1, 0, -1, -2)$ ($Ad = 0$, $c^\top d = 1 > 0$). Ratio test: $\min\{4/1, 5/2\} = 2.5$, giving $(3.5, 1, 1.5, 0)$, value 4.5. Support is now $\{1, 2, 3\}$, still dependent: $d = (-1, 2, -1, 0)$ has $Ad = 0$, $c^\top d = 1 > 0$; ratio test $\min\{3.5/1, 1.5/1\} = 1.5$ gives $(2, 4, 0, 0)$, the BFS of $\{1, 2\}$ with value 6, which is optimal.

Gotcha

- The theorem needs *both* feasibility and boundedness. Do not invoke it before checking them (or before an algorithm has established them).
- “Optimal solutions are vertices” is false as a universal statement; “some optimal solution is a vertex” is the theorem.
- Enumerating bases of the *canonical* form makes no sense; bases live in the *equational* form with slacks included.

Summary: The fundamental theorem

- Purification (Lemma 2.5.1): from any feasible point, walk along a null direction of A with the ratio test to a BFS at least as good; boundedness guarantees the walk stops.

- Fundamental theorem (Theorem 2.5.2): feasible and bounded above $\Rightarrow$ an optimal BFS exists.
- Consequence: comparing the finitely many BFSs solves the LP; large instances need a smarter walk (the simplex method).

Notes on the original slides

All row reductions, determinants and basic solutions in the Lecture 2 slides were recomputed and are correct (the three pivots, the six bases of the network example with their signs, $\det A_B = -1$ and -2 , the point $(6, 0, 0, -4)$). The following points deserve a clarification.

- **“The slack columns form an identity matrix” (slide 22).** This is true of the *original* matrix A , where the slack columns are e_1, e_2, e_3 , and it is what guarantees $\text{rank } A = 3$ in advance. It is *not* true of the row-reduced matrix shown on that slide, whose slack columns are $(1, -2, 1)$, $(0, 1, -2)$, $(0, 0, 1)$.
- **“The fourth inequality is also redundant” (slide 19).** True, but not by simply adding constraints: one needs $x \geq 0$ as well. Indeed $\frac{1}{2}(\text{first}) + \frac{3}{2}(\text{second})$ gives $3.5x_1 + 5x_2 + 2x_3 \leq 17$, which dominates $3x_1 + 5x_2 + x_3 \leq 18$ only because $x_1, x_3 \geq 0$.
- **“For equational form, vertices are exactly the BFSs” (slide 32).** This needs A to have full row rank (redundant equations removed), as the slides assume from slide 33 on; Theorem 2.4.10 states the hypothesis explicitly.
- **Theorem on slide 41.** The statement “feasible set nonempty and objective bounded above $\Rightarrow$ an optimal BFS exists” silently includes the claim that an optimum exists at all; the slides do not prove it. Lemma 2.5.1 and Theorem 2.5.2 supply the missing argument.
- **Basic solution of the first choice (slide 24).** The slide says “this choice gives no solution,” which is right; note that this does not mean the LP is infeasible (it is not), nor that x_1, x_2, x_3 can never all be positive in a feasible point (e.g. $(1, 1, 1)$ is feasible). It only means these three columns cannot form a basis.
- **Bases versus corners (slide 39).** “Four feasible bases, four vertices” holds here because no BFS is degenerate. In general several bases can give the same vertex (see the degeneracy remark after Proposition 2.4.4).

Chapter takeaways

Summary: The geometry and algebra of linear programs

- **Canonical form** $\max c^\top x$, $Ax \leq b$, $x \geq 0$ is the modelling format; **equational form** $\max c^\top z$, $Az = b$, $z \geq 0$ is the computing format. Slacks (for $\leq$), surpluses (for $\geq$) and variable shifts translate one into the other exactly.
- **Linear algebra** organises the equalities: row-reduce $[A \mid b]$, detect in-

consistency, delete zero rows, reach full row rank.

- **A basis** is m independent columns; its basic solution sets the other variables to 0; it is a **BFS** if nonnegative. BFSs are exactly the **vertices** of the feasible set, and there are finitely many.
- **The fundamental theorem:** a feasible, bounded LP has an optimal BFS. Hence comparing vertices suffices, and the stage is set for the simplex method.